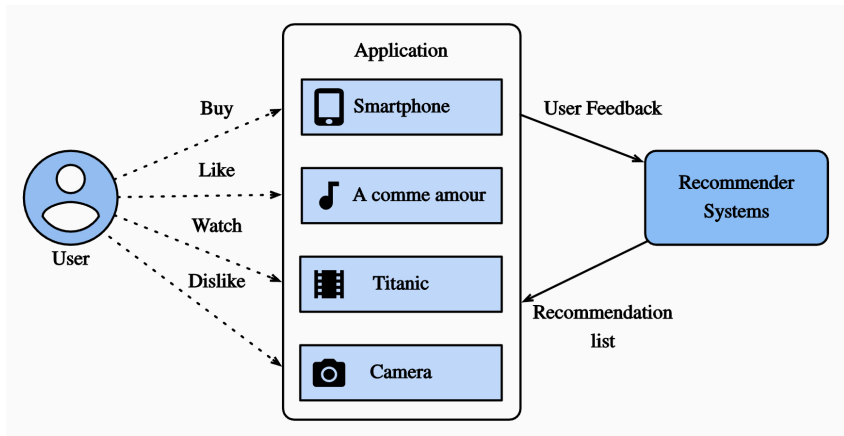


PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

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Recommender Systems



(source)

Recommendations: Goals and Challenges

Goals:

1. Maximize **engagement** / **conversion rates**
2. Balance **exploitation** (accuracy) and **exploration** (novelty)

Challenges:

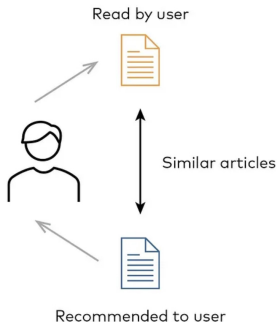
1. **Personalization**
2. **“Cold Starts”**
3. **Interpretability**
4. **Scalability** (“needle in a haystack”)

Content-Based Systems

Content-Based Filtering:

- Relies on **item metadata**
- Often uses **unsupervised learning** (patterns / similarity)
- Strengths: accuracy, new users / items, transparency, scalability
- Weaknesses: novelty / diversity, data demands
- Applications: search systems, niche platforms (bookstores, etc.)

CONTENT-BASED FILTERING

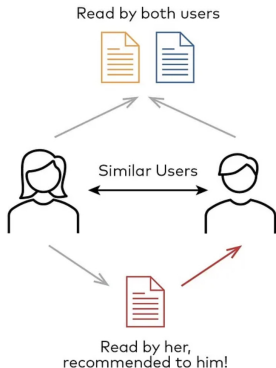


Collaborative Systems

Collaborative Filtering:

- Relies on **user traits + feedback** (explicit + implicit)
- May be **item-based** or **user-based**
- Strengths: balance accuracy with novelty; lower data demands
- Weaknesses: struggles with new items/users; less interpretable
- Applications: platforms with high engagement (e.g., YouTube, NetFlix, etc.)

COLLABORATIVE FILTERING



Systems, Compared

Example Goal: Recommend The Next Song

	System Type	
	<i>Content-Based</i>	<i>Collaborative</i>
Data: user choice, plus Intuition	Song Metadata Recommend a “similar” song	Other Users' Traits + Choices Recommend a song liked / listened to by “similar” users
Model	Unsupervised (e.g., clustering)	Supervised (predictive)
Explanation	“I recommended this because it's similar to other thing(s) you have liked before.”	“I recommended this because people who are like you seem to like it.”
Cold start	Easy; relies on item-level data	Difficult; user “similarity” is ill-defined

Hybrid systems:

- ...combine content-based and collaborative filters
- ...attempt to overcome the drawbacks of each
- E.g. (common): content-based recommendations for new / less-active users, collaborative recommendations for older / active users
- Many use *reinforcement learning* (real-time adjustments to algorithms based on recommendation “success”)

Other Recommenders

Session-based systems: Rely on data within a given user session.

Multi-criteria systems: Optimize for multiple user preferences (e.g., movie traits)

Context-aware systems:

- Context = location, time, surroundings, events, etc.
- Greater personalization, data demands
- E.g., Uber/Lyft route recommenders

Generative systems: More on these later...

Smith (2021):

- Goals: “retention” and “time spent”
- Model: Hybrid (user + item data), reinforcement learning
- Simplified algorithm:

$$\begin{aligned} \text{Pr(Recommend)} = & f[\text{Pr(Play)} \times V(\text{Play}) + \text{Pr(Like)} \times V(\text{Like}) + \\ & \text{Pr(Comment)} \times V(\text{Comment}) + E(\text{Playtime}) \times V(\text{Playtime})] \end{aligned}$$

- System combats gaming (“like bait”) and boredom
- Advantage: “fantastic volumes of data, highly engaged users, and a setting where users are amenable to consuming algorithmically recommended content”

(source)

Things to remember:

- **Data**
 - Every system constantly collects data
 - Richer + more continuous signal → better recommendations
- **User Profiles:** User data are key to personalized recommendations
- **Accuracy vs. Novelty**
 - Tradeoffs → algorithm
 - Balance of user vs. others' behavior
 - Platform-specific (via users' risk acceptance)